SINGLE PEAKS IN TRUNCATED STIELTJES INTEGRAL EQUATIONS

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Abstract. We study maxima of curves defined by normalized, truncated Stieltjes integrals. For a positive profile h, strict decrease of its elasticity $xh'(x)/h(x)$ on its declining side gives a unique nondegenerate maximum at every level for which the curve exists. This proves a uniqueness conjecture of Viakhirev and makes its additional existence hypotheses unnecessary. We treat every positive power of the kernel, on a finite interval or the positive half-line: all positive levels occur for powers at least one, whereas a uniquely maximized boundary integral gives the exact level threshold for smaller powers. We determine the movement and limiting positions of the peak. For log-normal densities, the normalized polynomial cutoffs proposed by Viakhirev have maximizing locations and heights converging at rate $O(R^{-1})$ to limits independent of the cutoff exponent. In contrast, smooth strictly unimodal profiles can produce arbitrarily many critical points.

1. Introduction

A moving upper limit can change the shape of an implicitly defined function even when the integrand has a single peak. Consider a probability density ρ and the equation

$$\int_0^m \frac{x\rho(x)}{C(m)-x}\,\mathrm{d}x = 1, \qquad C(m) > m. \tag{1}$$

The quantity of interest is the dimensionless ratio $\beta(m) = C(m)/m$. Viakhirev derives its derivative and proves the existence of a critical point under an additional sign-change hypothesis. His Conjecture 10 asserts that this point is unique when $h(x) = x\rho(x)$ is positive and strictly log-concave [6]. The same paper asks whether the maximizing cutoffs obtained by polynomially truncating a log-normal density converge as the truncation range increases [6].

We prove the conjecture and the cutoff limit. The main observation is a boundary identity for the derivative of a dilation average. After division by $h(m)$, its sign is controlled by an average of the ratios $h(mt)/h(m)$, $0 < t < 1$. These ratios increase strictly after the mode under a one-sided condition on the elasticity $xh'(x)/h(x)$. Every stationary point is consequently a strict downward crossing of the derivative. Treating the maximum as a function of the kernel parameter then resolves the implicit equation at every positive level.

This argument also determines the position of the peak. It lies strictly between the mode of h and the maximizer of its running average; for powers at least one, both bounds are attained as limits when the prescribed level varies. For powers between zero and one, an integrable boundary kernel instead determines the largest admissible level and the limiting peak at that level. The construction extends to the half-line, where it applies to log-normal profiles although their ordinary logarithm is eventually convex. We conclude with an explicit three-critical-point example and then a family with arbitrarily many critical points, all with a strictly unimodal profile. These constructions show that a single mode imposes no finite bound on the critical-point count.

There is a classical connection between concavity and preservation of unimodality. Ibragimov characterizes densities whose convolution with every unimodal density is unimodal by log-concavity [2]. The multiplicative counterpart of Cuculescu and Theodorescu is recalled by Simon in terms of concavity of $s \mapsto \log f(e^s)$ [5]. In logarithmic coordinates, our fixed-parameter integral is also a convolution with a one-sided decreasing kernel. The arguments below use only the declining side of

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the profile and retain the strict derivative information needed for the implicit curve, the movement of its peak, and the cutoff limit.

Parameter derivatives of positive integrals also admit probabilistic sign criteria. Derbazi develops such criteria through normalized weights and covariance identities, including applications to truncated integrals and Stieltjes-transform quotients [1]. Our parameter comparisons use a related subtraction at a single sign change. The specific boundary identity proved below controls dilation of the profile with no monotonicity assumption on its elasticity before the mode; it leads to the implicit peak theorem and its two level-existence regimes.

2. Profiles and the main theorem

Let $L \in (0, \infty]$. Throughout, a profile h is a real function in $C^1([0, L))$, positive on $(0, L)$, such that

$$h(0) = 0, \qquad \lim_{x \uparrow L} h(x) = 0. \tag{2}$$

When $L = \infty$, the second limit means $x \to \infty$. Suppose that h is strictly increasing up to a point $a \in (0, L)$ and strictly decreasing afterwards. Its elasticity is

$$g(x) = \frac{x h'(x)}{h(x)}, \qquad 0 < x < L. \tag{3}$$

Our additional assumption is that g is strictly decreasing on (a, L). No monotonicity of g is required before a. In particular, $g(x) < 0$ for $x > a$: differentiability gives $g(a) = 0$, and strict decrease after a makes it negative there.

For $p > 0$, $b > 1$ and $0 < m < L$, put

$$T_p(m, b) = \int_0^1 \frac{h(mt)}{(b - t)^p} \, \mathrm{d}t, \qquad A(m) = \int_0^1 h(mt) \, \mathrm{d}t = \frac{1}{m} \int_0^m h(x) \, \mathrm{d}x. \tag{4}$$

The same transform in the original coordinates is

$$T_p(m, b) = m^{p-1} \int_0^m \frac{h(x)}{(mb - x)^p} \, \mathrm{d}x. \tag{5}$$

Thus $p = 1$, $h = x\rho$ and level one recover (1) exactly.

Theorem 2.1. *Under the assumptions above, the following statements hold.*

(1) *The running average A has a unique critical point $b_h \in (a, L)$. It is its strict global maximum and satisfies*

$$A(b_h) = h(b_h). \tag{6}$$

(2) *For each $p > 0$ and $b > 1$, the function $m \mapsto T_p(m, b)$ has a unique critical point $m_p(b)$, with*

$$a < m_p(b) < b_h, \qquad (T_p)_{mm}(m_p(b), b) < 0. \tag{7}$$

It is a strict global maximum.

(3) *For each $p \geq 1$ and $\lambda > 0$, the equation*

$$T_p(m, \beta_{p,\lambda}(m)) = \lambda, \qquad \beta_{p,\lambda}(m) > 1, \tag{8}$$

defines a C^2 function on all of $(0, L)$. It has exactly one critical point $m^*_{p,\lambda}$. This point is a nondegenerate strict global maximum, and

$$\beta'_{p,\lambda}(m) > 0 \quad (0 < m < m^*_{p,\lambda}), \qquad \beta'_{p,\lambda}(m) < 0 \quad (m^*_{p,\lambda} < m < L). \tag{9}$$

The use of b_h for the running-average maximizer is separate from the parameter b of the kernel. The inequality (7) is independent of that parameter and will also control limits on the half-line.

Corollary 2.2. *Let $M < \infty$ and let $\rho \in C^1([0, M])$ be a probability density with $\rho(M) = 0$. If $h(x) = x\rho(x)$ is positive and strictly log-concave on $(0, M)$, then (1) has a unique solution $C(m) > m$ for every $0 < m < M$. The ratio $C(m)/m$ has exactly one critical point, a nondegenerate strict global maximum. In particular, Viakhirev's Conjecture 10 holds, and its extra sign-change and domain hypotheses are automatic.*

Proof. The continuous function h vanishes at both endpoints. Strict log-concavity gives a unique mode and strict increase and decrease on its two sides. On the declining side, $s = h'/h$ is negative and nonincreasing. If $a < x < y < M$, then

$$y\, s(y) \le y\, s(x) < x\, s(x). \tag{10}$$

Hence the elasticity is strictly decreasing there. Theorem 2.1 applies with $p = \lambda = 1$. Positivity gives the full domain, and the negative derivative after the maximum supplies the additional sign-change point. □

Only log-concavity on the declining side was used in (10). Another sufficient condition is strict concavity of $u \mapsto \log h(e^u)$, since its derivative is $g(e^u)$. These sufficient conditions are distinct; the latter includes the log-normal case treated below.

Example 2.3. *For $h(x) = x(1 - x)$ on $[0, 1]$, the mode is $a = 1/2$ and*

$$A(m) = \frac{m}{2} - \frac{m^2}{3}, \qquad b_h = \frac{3}{4}. \tag{11}$$

Every peak covered by Theorem 2.1 therefore lies strictly between $1/2$ and $3/4$. The limiting assertions of Theorem 4.1 below show that both bounds are sharp. Multiplication of h by a positive constant changes the effective level, but leaves a, b_h and $m_p(b)$ unchanged.

3. A boundary ratio and the proof of uniqueness

We first isolate the derivative argument for a fixed weight. Let w be a positive C^1 function on $[0, 1]$, and assume that

$$q(t) = w(t) + tw'(t) \ge 0 \qquad (0 \le t \le 1). \tag{12}$$

Since $q(0) = w(0) > 0$, this is a nonzero weight. Define

$$U(m) = \int_0^1 h(mt)w(t)\,\mathrm{d}t, \qquad R_w(m) = \frac{1}{w(1)} \int_0^1 \frac{h(mt)}{h(m)} q(t)\,\mathrm{d}t. \tag{13}$$

Lemma 3.1. *For $0 < m < L$,*

$$mU'(m) = w(1)h(m)(1 - R_w(m)). \tag{14}$$

For $m > a$, $R_w'(m) > 0$. Also $U'(m) > 0$ for $0 < m \le a$. Consequently U has exactly one critical point, a nondegenerate strict global maximum in (a, L).

Proof. Differentiation on compact subintervals of $(0, L)$ and integration by parts in t give

$$m\,U'(m) = \int_0^1 t \frac{\mathrm{d}}{\mathrm{d}t} h(m\,t)w(t)\,\mathrm{d}t = h(m)w(1) - \int_0^1 h(m\,t)(w(t) + t\,w'(t))\,\mathrm{d}t. \tag{15}$$

Here $\mathrm{d}h(mt)/\mathrm{d}t = mh'(mt)$, and the boundary term at zero vanishes. This proves (14).

For $m > a$ and $0 < t < 1$,

$$\frac{\partial}{\partial m} \frac{h(m\,t)}{h(m)} = \frac{h(m\,t)}{m\,h(m)} (g(m\,t) - g(m)) > 0. \tag{16}$$

If $mt \le a$, the first elasticity is nonnegative and the second is negative. If $mt > a$, strict decrease of g gives the same conclusion. Differentiation of (13) is justified by the boundedness of h and h' on

each compact interval $[0, m_1]$, with $h(m)$ bounded away from zero on the interval of differentiation. Positivity in (16) and $q > 0$ near zero imply $R'_w(m) > 0$.

For $m \le a$, $h' \ge 0$ on $[0, m]$ and is positive on some interior interval, because h is strictly increasing. Thus $U'(m) = \int_0^1 t h'(mt) w(t)\,\mathrm{d}t > 0$. After a, (14) has at most one zero, with a positive sign before it and a negative sign after it. The identity itself shows that U' is continuously differentiable, even though h need not be twice differentiable. At a zero m_0,

$$U''(m_0) = -\frac{w(1)h(m_0)}{m_0} R'_w(m_0) < 0. \tag{17}$$

It remains to show that a critical point exists. If $L < \infty$, choose a closed interval inside $(0, 1)$ on which q is positive. The values $h(mt)$ on this interval are bounded away from zero as $m \uparrow L$, whereas $h(m) \to 0$. Thus $R_w(m) \to \infty$, and (14) is eventually negative. If $L = \infty$, boundedness of h and dominated convergence give $U(m) \to 0$. Since $U(a) > 0$, the mean value theorem gives a point after a where $U' < 0$. In both cases continuity supplies a zero. The sign statements prove the global assertion. □

The measure with density $q(t)/w(1)$ is a probability measure, since its integral is $[tw(t)]_0^1/w(1) = 1$. This normalization is useful because a critical point is precisely the condition that the mean of $h(mt)/h(m)$ under this measure is one.

Proof of Theorem 2.1, parts (1) and (2). Take $w = 1$. Then $U = A$ and $R_w = A/h$. Lemma 3.1 gives a unique critical point $b_h \in (a, L)$, and (14) at that point proves (6).

For $w(t) = (b - t)^{-p}$, the probability density in the lemma is

$$r_{b,p}(t) = \frac{(b-1)^p (b + (p-1)t)}{(b-t)^{p+1}}. \tag{18}$$

It is positive and strictly increasing in t for every $p > 0$: its derivative has the sign of $p(2b+(p-1)t) > 0$. At $m = b_h$, the function

$$v(t) = \frac{h(b_h t)}{h(b_h)} - 1 \tag{19}$$

has exactly one sign change in $(0, 1)$, from negative to positive. This follows from strict increase up to a and strict decrease thereafter. Its unweighted integral is zero by (6). If c is its crossing point, then

$$\int_0^1 v(t) r_{b,p}(t)\,\mathrm{d}t = \int_0^1 v(t)\big(r_{b,p}(t) - r_{b,p}(c)\big)\,\mathrm{d}t > 0. \tag{20}$$

Thus $(T_p)_m(b_h, b) < 0$, whereas $(T_p)_m(a, b) > 0$. Lemma 3.1 proves part (2), including the strict bounds and nondegeneracy. □

Proof of Theorem 2.1, part (3). For each fixed $m \in (0, L)$, $T_p(m, b)$ is continuous and strictly decreasing in b, with

$$(T_p)_b(m, b) = -p \int_0^1 \frac{h(mt)}{(b-t)^{p+1}}\,\mathrm{d}t < 0. \tag{21}$$

It tends to zero as $b \to \infty$. As $b \downarrow 1$, it tends to infinity because $h(m) > 0$ and $p \ge 1$. This proves the existence and uniqueness of the solution in (8). The transform is C^2 in (m, b): the first m-derivative has the boundary expression (15), whose derivative uses only h', while derivatives in b are dominated on compact sets. The implicit function theorem gives $\beta_{p,\lambda} \in C^2$.

Let

$$E_p(b) = T_p(m_p(b), b). \tag{22}$$

The equation $R_w(m_p(b)) = 1$ and $R'_w > 0$ show that m_p is C^1. At its maximum,

$$E'_p(b) = (T_p)_b(m_p(b), b) < 0. \tag{23}$$

Moreover, $E_p(b) \to \infty$ as $b \downarrow 1$, since it is at least $T_p(a, b)$, and

$$0 < E_p(b) \le \frac{\|h\|_\infty}{(b-1)^p} \longrightarrow 0 \qquad (b \to \infty). \tag{24}$$

Therefore $E_p(b_*) = \lambda$ has exactly one solution.

Differentiating (8) shows that a critical point of $\beta_{p,\lambda}$ must have $m = m_p(b)$ and $E_p(b) = \lambda$. There is consequently exactly one such point, namely $m_p(b_*)$. For all m, $T_p(m, b_*) \le \lambda$; strict decrease in b gives $\beta_{p,\lambda}(m) \le b_*$, with equality only at that point. At the critical point,

$$\beta''_{p,\lambda} = -\frac{(T_p)_{mm}}{(T_p)_b} < 0. \tag{25}$$

The derivative is positive for $m \le a$ by Lemma 3.1. Its unique zero and its strict downward crossing give (9) on the entire domain. □

4. Movement and limiting position of the peak

The same single-crossing calculation controls the kernel parameters. We state the dependence at fixed b separately from dependence at a prescribed level.

Theorem 4.1. *For fixed $p \ge 1$, $m_p(b)$ is strictly increasing in b, and*

$$\lim_{b\downarrow 1} m_p(b) = a, \qquad \lim_{b\to\infty} m_p(b) = b_h. \tag{26}$$

For fixed $b > 1$, $m_p(b)$ is strictly decreasing in $p \ge 1$ and tends to a as $p \to \infty$. For fixed p, the implicit peak location $m^*_{p,\lambda}$ is strictly decreasing in λ, with

$$\lim_{\lambda\to\infty} m^*_{p,\lambda} = a, \qquad \lim_{\lambda\downarrow 0} m^*_{p,\lambda} = b_h. \tag{27}$$

Proof. Write $R(m, b, p) = \int_0^1 h(mt) r_{b,p}(t)/h(m)\,\mathrm{d}t$. At a root $R = 1$, the function $v(t) = h(mt)/h(m) - 1$ has one negative-to-positive crossing c. Since $r_{b,p}$ has integral one, differentiation in a parameter θ gives

$$R_\theta = \int_0^1 v(t) r_{b,p}(t)\, \partial_\theta \log r_{b,p}(t)\,\mathrm{d}t. \tag{28}$$

For the parameter b,

$$\partial_b \log r_{b,p}(t) = \frac{p}{b-1} + \frac{1}{b + (p-1)t} - \frac{p+1}{b-t}. \tag{29}$$

Its derivative in t is

$$-\frac{p-1}{(b+(p-1)t)^2} - \frac{p+1}{(b-t)^2} < 0. \tag{30}$$

Subtracting the score at c in (28) proves $R_b < 0$. Because $R_m > 0$, implicit differentiation yields $m'_p(b) = -R_b/R_m > 0$.

For the exponent,

$$\partial_p \log r_{b,p}(t) = \log(b-1) - \log(b-t) + \frac{t}{b + (p-1)t}. \tag{31}$$

Its derivative in t is $1/(b-t) + b/(b + (p-1)t)^2 > 0$. The same subtraction now proves $R_p > 0$, hence strict decrease of $m_p(b)$ in p.

All peak locations lie in $[a, b_h]$, a compact subset of $(0, L)$. As $b \to \infty$, $b^p(b-t)^{-p} \to 1$ uniformly on $[0, 1]$. Thus $b^p T_p(m, b) \to A(m)$ uniformly for $m \in [a, b_h]$. Uniqueness of the maximizer of A gives the second limit in (26).

Let $J_p(b) = \int_0^1 (b-t)^{-p}\,\mathrm{d}t$. For $p \geq 1$, $J_p(b) \to \infty$ as $b \downarrow 1$, while the integral over $[0, 1-\delta]$ remains bounded for each $\delta > 0$. Hence the normalized weight $(b-t)^{-p}/J_p(b)$ concentrates at $t = 1$. Uniform continuity of $h(mt)$ on $[a, b_h] \times [0, 1]$ gives

$$\frac{T_p(m, b)}{J_p(b)} \longrightarrow h(m) \quad \text{uniformly for } m \in [a, b_h]. \tag{32}$$

The unique mode of h gives the first limit. With b fixed, the normalized weights also concentrate at 1 as $p \to \infty$: the mass below $1-\delta$ is exponentially small compared with that on $[1-\delta/2, 1]$. This proves the exponent limit. Finally, E_p is strictly decreasing from infinity to zero, so the solution $b_*(\lambda)$ is strictly decreasing with the corresponding endpoint limits. Since $m^*_{p,\lambda} = m_p(b_*(\lambda))$, (27) follows. □

The restriction $p \geq 1$ distinguishes two existence regimes. For a smaller exponent the transform remains finite at $b = 1$, and this boundary transform determines precisely which levels have solutions.

Theorem 4.2. *Let $0 < p < 1$ and define*

$$S_p(m) = \int_0^1 \frac{h(mt)}{(1-t)^p}\,\mathrm{d}t. \tag{33}$$

This function has a unique critical point $c_p \in (a, b_h)$ and is strictly increasing before it and strictly decreasing afterwards. Put $\lambda_c = S_p(c_p)$. The level equation (8) has a nonempty domain precisely for $0 < \lambda < \lambda_c$. In that case its domain is the interval

$$\Omega_{p,\lambda} = \{m \in (0, L) : S_p(m) > \lambda\}. \tag{34}$$

On this interval, the solution is C^2 and has exactly one critical point, a nondegenerate strict global maximum. Its location decreases strictly with λ, with

$$\lim_{\lambda \uparrow \lambda_c} m^*_{p,\lambda} = c_p, \qquad \lim_{\lambda \downarrow 0} m^*_{p,\lambda} = b_h. \tag{35}$$

For fixed $0 < p < 1$, $m_p(b)$ increases strictly with b, from c_p to b_h. For each fixed $b > 1$, $m_p(b)$ decreases strictly over all $p > 0$, with limits b_h at $p \downarrow 0$ and a at $p \to \infty$. On $0 < p < 1$, c_p decreases strictly from b_h to a, whereas λ_c increases strictly from $A(b_h)$ to infinity.

Proof. The integrability of $(1-t)^{-p}$ and the local boundedness of h' show that S_p is C^1. Let $w(t) = (1-t)^{-p}$ and $q(t) = (1 + (p-1)t)/(1-t)^{p+1}$. Integrating by parts up to $u < 1$, and subtracting $h(m)$ inside the integral, gives

$$\frac{m S_p'(m)}{h(m)} = \int_0^1 \left(1 - \frac{h(mt)}{h(m)}\right) \frac{1 + (p-1)t}{(1-t)^{p+1}}\,\mathrm{d}t. \tag{36}$$

Indeed, the discarded boundary term is $u(h(mu) - h(m))w(u)/h(m) = O((1-u)^{1-p})$. The integral converges, because $h(mt) - h(m) = O(1-t)$ near 1. For $m > a$, its right side is strictly decreasing in m by (16); this comparison uses convergent integrals and requires no derivative of the singular integral.

The derivative S_p' is positive for $m \leq a$. At $m = b_h$, the function $t \mapsto t h'(b_h t)$ is nonnegative before a/b_h, negative after it, and has integral $A'(b_h) = 0$. Multiplying it by the strictly increasing weight $(1-t)^{-p}$ makes its integral strictly negative: subtract the weight at a/b_h and compare the signs on the two sides. Thus $S_p'(b_h) < 0$. Equation (36) proves the existence and uniqueness of c_p, and the signs on both sides.

For every fixed m, $T_p(m,b)$ decreases strictly from $S_p(m)$ to zero as b runs from 1 to infinity. This proves (34). Strict unimodality of S_p proves that this set is an interval and that it is nonempty precisely below λ_c.

For $0 < p < 1$, the derivative in (30) is still negative: $b + (p-1)t \geq b - t > 0$ and $1 - p < 1 + p$ show that its positive first term is smaller in magnitude than its negative second term. The proof of $R_b < 0$ and $m_p'(b) > 0$ therefore applies. The derivative in t of (31) remains positive for all $p > 0$, giving strict decrease in p on this entire range. As $p \downarrow 0$, the kernels converge uniformly to 1 for fixed b, and the compact peak bound gives $m_p(b) \to b_h$.

As $b \downarrow 1$, dominated convergence gives $T_p(m,b) \to S_p(m)$ uniformly for $m \in [a, b_h]$. Uniqueness of the maximizer of S_p gives $m_p(b) \to c_p$ and $E_p(b) \to \lambda_c$. As $b \to \infty$, the uniform running-average argument in Theorem 4.1 gives $m_p(b) \to b_h$ and $E_p(b) \to 0$. Thus the strictly decreasing envelope attains each $\lambda \in (0, \lambda_c)$ exactly once. The proof of part (3) of Theorem 2.1, restricted to (34), gives the unique nondegenerate maximum. The envelope and peak monotonicities give (35). The limit as $p \to \infty$ was proved in Theorem 4.1.

For $0 < p < q < 1$, the function $th'(c_p t)/(1-t)^p$ has integral zero and changes sign from nonnegative to negative at a/c_p. Multiplication by the strictly increasing factor $(1-t)^{-(q-p)}$ makes its integral negative, so $S_q'(c_p) < 0$ and $c_q < c_p$. As $p \downarrow 0$, $S_p \to A$ uniformly on $[a, b_h]$ by domination with any fixed integrable exponent; hence $c_p \to b_h$ and $\lambda_c \to A(b_h)$. As $p \uparrow 1$, the probability weights $(1-p)(1-t)^{-p}$ concentrate at 1, giving $(1-p)S_p \to h$ uniformly on this same interval. Thus $c_p \to a$ and $\lambda_c \to \infty$. Strict increase of λ_c follows from the pointwise strict increase of $S_p(m)$ in p. □

5. Log-normal profiles and polynomial cutoffs

Let $\mu \in \mathbb{R}$, $\sigma > 0$, and let

$$\rho(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\log x - \mu)^2}{2\sigma^2}\right), \qquad x > 0. \tag{37}$$

For $h = x\rho$, the elasticity is

$$g(x) = -\frac{\log x - \mu}{\sigma^2}. \tag{38}$$

The profile extends smoothly by zero at the origin, tends to zero at infinity, and has mode e^{μ}. Theorem 2.1 applies on the half-line. In contrast, $(\log h)''(x) = (\log x - \mu - 1)/(\sigma^2 x^2)$ is positive for sufficiently large x.

For $R > 0$ and $\alpha > 1$, define the exact normalized cutoff density

$$\rho_{R,\alpha}(x) = \frac{\rho(x)(R-x)^{\alpha-1}}{Z_{R,\alpha}} \quad (0 < x < R), \qquad Z_{R,\alpha} = \int_0^R \rho(x)(R-x)^{\alpha-1}\,\mathrm{d}x. \tag{39}$$

It is zero outside $[0, R]$. These are the regularizations proposed in Section 6.3 of Viakhirev's paper [6].

Theorem 5.1. *Fix $p > 0$, $\lambda > 0$, $\mu \in \mathbb{R}$, $\sigma > 0$ and $\alpha > 1$. If $0 < p < 1$, assume also $\lambda < \lambda_c$ for the untruncated profile, with λ_c as in Theorem 4.2. For all sufficiently large R, the implicit curve for $h_{R,\alpha}(x) = x\rho_{R,\alpha}(x)$ has a nonempty domain and a unique maximum. Let $m^*_{R,\alpha}$ and $B^*_{R,\alpha}$ be its location and height, and let m^* and B^* denote the corresponding quantities for the untruncated density (37). Then*

$$|m^*_{R,\alpha} - m^*| + |B^*_{R,\alpha} - B^*| = O(R^{-1}) \qquad (R \to \infty). \tag{40}$$

The limiting location and height are independent of α. More precisely, write $T = T_p$ for the untruncated transform, put $\mu_1 = e^{\mu+\sigma^2/2}$, and define

$$V(m,b) = \int_0^1 \frac{mt\,h(mt)}{(b-t)^p}\,\mathrm{d}t. \tag{41}$$

With all quantities on the right evaluated at (m^*, B^*), put

$$v = \frac{V - \mu_1\lambda}{T_b}, \qquad u = \frac{V_m - T_{mb}v}{T_{mm}}. \tag{42}$$

Then

$$(m^*_{R,\alpha}, B^*_{R,\alpha}) = (m^*, B^*) + \frac{\alpha-1}{R}(u,v) + O(R^{-2}). \tag{43}$$

Thus the first-order displacement is linear in $\alpha-1$, with coefficients independent of α. The implicit constants may depend on all fixed parameters. In particular, at $p = \lambda = 1$ this establishes the limit asked for in the source paper.

Proof. On $(0,R)$, multiplication by the cutoff changes the elasticity to

$$g_{R,\alpha}(x) = -\frac{\log x - \mu}{\sigma^2} - (\alpha-1)\frac{x}{R-x}. \tag{44}$$

This is strictly decreasing from infinity to minus infinity. Hence the cutoff profile has a unique mode, is strictly increasing before it and strictly decreasing afterwards, and satisfies the hypotheses of Theorem 2.1. Only continuity is needed at R; the derivative may be unbounded there if $1 < \alpha < 2$.

Put $c = \alpha - 1$ and $q_R(x) = (1 - x/R)_+^c$, where $u_+ = \max(u, 0)$. The factor R^c cancels in (39), so

$$h_{R,\alpha}(x) = \frac{h(x)q_R(x)}{N_R}, \qquad N_R = \int_0^\infty \rho(x)q_R(x)\,\mathrm{d}x. \tag{45}$$

For $x \geq 0$,

$$0 \leq 1 - q_R(x) \leq \max(1,c)\frac{x}{R}. \tag{46}$$

For $0 \leq x/R \leq 1$, use the mean value theorem when $c \geq 1$ and $(1-u)^c \geq 1-u$ when $0 < c \leq 1$; for $x/R \geq 1$ the bound is immediate. The log-normal law has finite first moment, as follows by completing the square after the substitution $u = \log x$. Therefore $N_R = 1 + O(R^{-1})$. On every fixed $[0,K]$, $q_R = 1 + O(R^{-1})$ and $q_R' = O(R^{-1})$ uniformly for $R > 2K$. It follows that

$$\|h_{R,\alpha} - h\|_{C^1([0,K])} = O(R^{-1}). \tag{47}$$

We first establish eventual existence and convergence of the actual maxima. Fix m in the domain of the untruncated curve, and choose two values of b, both greater than one, on opposite sides of its solution. Strict decrease in b gives opposite strict signs relative to λ at these two values. By (47), the same signs hold for the cutoff transform for large R. Its solution therefore exists there and converges to the untruncated solution. The derivative formula obtained from (8) then implies convergence of its derivative. In the case $0 < p < 1$, this argument also proves that the cutoff domain is eventually nonempty, so Theorem 4.2 gives its unique maximum. For $p \geq 1$, Theorem 2.1 already gives existence for every R.

Choose $m_- < m^* < m_+$ sufficiently close to m^* that both lie in the untruncated domain. The untruncated derivatives at these two cutoffs are strictly positive and strictly negative. The same is true for the cutoff profiles for all sufficiently large R. Their unique maximizing locations must lie between m_- and m_+. Making the two cutoffs approach m^* proves $m^*_{R,\alpha} \to m^*$. Local convergence of the transforms and strictness in b give $B^*_{R,\alpha} \to B^*$.

For the rate, write $T = T_p$ for the untruncated transform and consider the map

$$F(m,b) = \big(T(m,b) - \lambda,\ T_m(m,b)\big). \tag{48}$$

At (m^*, B^*), its Jacobian determinant is

$$-T_b(m^*, B^*)T_{mm}(m^*, B^*) \neq 0. \tag{49}$$

It has a continuously differentiable local inverse, which is Lipschitz on a sufficiently small closed neighborhood. Let F_R be the corresponding map for $h_{R,\alpha}$. The profile estimate gives $F_R - F = O(R^{-1})$ uniformly on this neighborhood, including its first derivatives; one may use (15) for the second m-derivative. Since the actual critical pairs already converge and satisfy $F_R = 0$, their images under F are $O(R^{-1})$. The local inverse proves (40). None of these limits depends on α. □

First-order displacement in Theorem 5.1. Keep $c = \alpha - 1$ and q_R from the preceding proof. The cutoff factor satisfies the global bound

$$\left| q_R(x) - 1 + \frac{cx}{R} \right| \leq C_c \frac{x^2}{R^2} \qquad (x \geq 0). \tag{50}$$

For $x/R \leq 1/2$, this is Taylor's theorem on a fixed compact interval; for $x/R \geq 1/2$, use $0 \leq q_R \leq 1$ and bound $2 + cx/R$ by a constant times $(x/R)^2$. Completing the square gives $\int x\rho(x)\,\mathrm{d}x = \mu_1$ and a finite second moment. Hence $N_R = 1 - c\mu_1/R + O(R^{-2})$. Taylor expansion of q_R and its derivative on each fixed $[0, K]$ now gives

$$\left\| h_{R,\alpha}(x) - h(x) - \frac{c}{R}(\mu_1 - x)h(x) \right\|_{C^1([0,K])} = O(R^{-2}). \tag{51}$$

By linearity in the profile,

$$F_R = F + \frac{c}{R}(\mu_1 T - V,\ \mu_1 T_m - V_m) + O(R^{-2}) \tag{52}$$

uniformly on the fixed neighborhood of the limiting critical pair. The untruncated log-normal profile is smooth, so F is C^2 there. Write $\Delta m = m^*_{R,\alpha} - m^*$ and $\Delta b = B^*_{R,\alpha} - B^*$. Both are $O(R^{-1})$ by (40). Expanding (52) at (m^*, B^*), and using $T_m = 0$, gives

$$T_b \Delta b + \frac{c}{R}(\mu_1 \lambda - V) = O(R^{-2}),$$
$$T_{mm}\Delta m + T_{mb}\Delta b - \frac{c}{R}V_m = O(R^{-2}).$$

The nonzero derivatives T_b and T_{mm} allow us to solve these two equations, giving (42) and (43). □

This conclusion concerns the deterministic equation (1). It does not assert a stochastic limit theorem for preferential-attachment networks. The fixed finite interval used in the peak-trapping argument is also important: convergence of densities alone would not control maximizers that could escape to infinity.

6. A strictly unimodal profile with several critical points

The elasticity hypothesis supplies information beyond the presence of a single mode. Viakhirev leaves open whether a unimodal but non-log-concave h can produce several critical points [6]. The following example answers this question within the smooth, finite-interval setting of (1).

Theorem 6.1. *There is a C^1 probability density ρ on $[0, 501]$, positive on $(0, 501)$ and vanishing polynomially at 501, for which $h(x) = x\rho(x)$ is strictly increasing on $(0, 3/2)$ and strictly decreasing on $(3/2, 501)$, but the ratio $C(m)/m$ in (1) has at least three distinct critical points.*

Proof. We first obtain strict inequalities for a step profile and then smooth it while preserving both the inequalities and a unique mode. Put $d = 11/20$ and

$$h_0(x) = \begin{cases} 1, & 1 \leq x \leq 2, \\ d, & 2 < x \leq 200, \\ 0, & \text{otherwise}, \end{cases} \qquad N = \int_0^\infty \frac{h_0(x)}{x}\,\mathrm{d}x = \log 2 + d \log 100. \tag{53}$$

Choose $b > 1$ by

$$d \log \frac{b}{b-1} = N + \frac{1}{500}. \tag{54}$$

Since $(N + 1/500)/d > \log 200$, this gives $1 < b < 200/199 < 101/100$. Define

$$T_0(m, b) = \int_0^m \frac{h_0(x)}{mb - x}\, \mathrm{d}x. \tag{55}$$

For $2 \le m \le 200$, direct integration yields

$$T_0(m, b) = N + \frac{1}{500} + D(bm), \qquad D(x) = \frac{9}{20} \log \frac{x-1}{x-2} - \frac{11}{20} \log \frac{x}{x-1}. \tag{56}$$

For $z = 1/(x-1) \in (0, 1)$, the absolutely convergent logarithm series gives

$$D(x) = \sum_{k \ge 1} \frac{z^{2k}}{2k} - \frac{1}{10} \sum_{k \ge 0} \frac{z^{2k+1}}{2k+1}. \tag{57}$$

At $m = 1/2$, $T_0 = 0 < N$. At $m = 2$,

$$T_0(2, b) = \log \frac{2b-1}{2b-2} > \log \frac{b}{b-1} - \log 2 > N. \tag{58}$$

For the last inequality, note that $N > 2 \log 2$ and $(1/d - 1)N = (9/11)N > \log 2$. At $m = 10$, $10/91 < z < 1/9$, and (57) gives

$$D(10b) \le \frac{z^2}{2(1-z^2)} - \frac{z}{10} < \frac{1}{160} - \frac{1}{91} < -\frac{1}{500}. \tag{59}$$

Thus $T_0(10, b) < N$. At $m = 100$, $0 < z < 1/99$, so

$$D(100b) \ge -\frac{z}{10(1-z^2)} > -\frac{99}{98000} > -\frac{1}{500}, \tag{60}$$

and $T_0(100, b) > N$. Finally, since $h_0 \le 1$ and its support is contained in $[1, 200]$,

$$T_0(400, b) \le \int_1^{200} \frac{1}{400 - x}\, \mathrm{d}x = \log \frac{399}{200} < \log 2 < N. \tag{61}$$

We have therefore obtained alternating strict signs relative to N at five increasing cutoffs.

Let k be a nonnegative smooth function supported in $(-1, 1)$ with integral one. For $0 < \varepsilon < 1/4$, set $k_\varepsilon(x) = k(x/\varepsilon)/\varepsilon$ and, on $[0, 501]$, define

$$B(x) = x^2 \left(1 - \frac{x}{501}\right)^{666}, \qquad h_\varepsilon(x) = (k_\varepsilon * h_0)(x) + \varepsilon B(x). \tag{62}$$

Here the convolution uses the extension of h_0 by zero to the real line. Its derivative is

$$(k_\varepsilon * h_0)'(x) = k_\varepsilon(x-1) - \frac{9}{20} k_\varepsilon(x-2) - \frac{11}{20} k_\varepsilon(x-200). \tag{63}$$

The positive term is supported below $5/4$, and the two negative terms are supported above $7/4$. The function B has its unique mode at $2 \cdot 501/(666 + 2) = 3/2$; its derivative is strictly positive before that point and strictly negative afterwards. Hence h_ε is strictly increasing to $3/2$ and strictly decreasing thereafter. It is positive throughout $(0, 501)$.

Normalize by

$$N_\varepsilon = \int_0^{501} \frac{h_\varepsilon(x)}{x}\, \mathrm{d}x, \qquad \rho_\varepsilon(x) = \frac{h_\varepsilon(x)}{N_\varepsilon x} \quad (x > 0). \tag{64}$$

The convolution term is supported away from both endpoints. Near zero and near 501, the density is a positive constant times $x(1 - x/501)^{666}$. It therefore extends to a C^1 probability density on

the closed interval, with polynomial vanishing of order 666 at its right endpoint. Multiplication by the positive normalizing constant leaves the mode of $x\rho_\varepsilon(x)$ unchanged.

As $\varepsilon \downarrow 0$, the convolution converges to h_0 in L^1, and its support stays in $[3/4, 201]$. It follows that $N_\varepsilon \to N$. At each of the five fixed cutoffs, the denominator in (55) is bounded away from zero, so the corresponding transform of h_ε converges to T_0. For all sufficiently small positive ε, the inequalities (58)–(61) and the inequality at $1/2$ therefore retain their strict signs relative to N_ε.

Positivity of ρ_ε ensures that (1) defines a C^1 ratio $\beta_\varepsilon > 1$ throughout $(0, 501)$, by strict decrease of the integral in β and its divergence as $\beta \downarrow 1$. The sign of $\beta_\varepsilon(m) - b$ is the sign of the unnormalized transform of h_ε at (m, b) minus N_ε. At

$$m = \frac{1}{2},\ 2,\ 10,\ 100,\ 400 \tag{65}$$

these signs are respectively $-, +, -, +, -$. The intermediate value theorem gives four distinct crossings of level b, one in each interval between successive cutoffs. Rolle's theorem between successive crossings gives at least three distinct critical points. Choose any sufficiently small ε to obtain the required density. □

Theorem 6.1 concerns unimodality of $h = x\rho$, the profile appearing in the derivative of the original equation. It does not contradict Corollary 2.2: its profile cannot satisfy the stated log-concavity hypothesis. In fact, the number of critical points is unbounded within this strictly unimodal class.

Theorem 6.2. *For every integer $r \geq 1$, there are $M < \infty$ and a C^1 probability density ρ on $[0, M]$, positive in its interior and vanishing polynomially at M, such that $h = x\rho$ is strictly unimodal with mode $3/2$, but $C(m)/m$ in (1) has at least $2r + 1$ distinct critical points.*

Proof. Keep $d = 11/20$ and the function D from (56). Its derivative is

$$D'(x) = \frac{(2d-1)x - 2d}{x(x-1)(x-2)}. \tag{66}$$

Thus D increases strictly to zero on $(11, \infty)$, where it is negative. For $0 \leq j < r$, set

$$x_j = 20 \cdot 4^j, \qquad \delta_j = -\frac{D(x_j) + D(2x_j)}{2}. \tag{67}$$

The numbers δ_j are positive and strictly decreasing, with $\delta_j + D(x_j) < 0 < \delta_j + D(2x_j)$. Put $B_0 = 2$ and $B_j = 3x_{j-1}$ for $1 \leq j < r$.

Let J tend to infinity and put $b_J = (1 - e^{-J})^{-1}$ and $a_j = d + \delta_j/J$. Define a step profile H_J equal to 1 on $[1, 2]$, to a_j on $(B_j, B_{j+1}]$ for $j < r - 1$, and to a_{r-1} on $(B_{r-1}, L_J]$, with zero values elsewhere. Choose L_J by

$$\log 2 + \sum_{j=0}^{r-2} a_j \log \frac{B_{j+1}}{B_j} + a_{r-1} \log \frac{L_J}{B_{r-1}} = dJ. \tag{68}$$

The sum is empty if $r = 1$. For sufficiently large J, this gives $L_J > 2x_{r-1}$ and $1 > a_0 > \cdots > a_{r-1} > 0$. The left side is exactly $N_J = \int H_J(x)/x \, \mathrm{d}x$.

At a fixed m strictly inside plateau j, the transform of the base profile, equal to 1 on $[1, 2]$ and to d thereafter, is $dJ + D(mb_J)$. The correction on the current plateau is

$$\frac{\delta_j}{J} \int_{B_j}^{m} \frac{\mathrm{d}x}{mb_J - x} = \delta_j + \frac{\delta_j}{J} \log \frac{b_J - B_j/m}{b_J}. \tag{69}$$

Each earlier-plateau correction is $O(J^{-1})$, since its upper endpoint is strictly below m. Consequently

$$\int_0^m \frac{H_J(x)}{mb_J - x} \, \mathrm{d}x - N_J \longrightarrow \delta_j + D(m). \tag{70}$$

At x_j and $2x_j$ the limiting signs are respectively negative and positive. At $1/2$ the transform is zero. At 2 it is greater than $J - \log 2 > dJ$ for large J. At $2L_J$ it is at most $\log((2L_J - 1)/L_J) < \log 2 < dJ$. Fix J large enough that all these inequalities hold. The increasing list $1/2, 2, x_0, 2x_0, \ldots, x_{r-1}, 2x_{r-1}, 2L_J$ has alternating signs, starting and ending negative.

Choose an integer $q \geq 2$ such that $M = 3(q+2)/4 > 2L_J + 1$. Smooth H_J by the same mollifier as before and add $\varepsilon x^2(1-x/M)^q$. All downward jumps of H_J lie at or beyond 2, and its sole upward jump is at 1. The added function has its unique mode at $2M/(q+2) = 3/2$. For $0 < \varepsilon < 1/4$, the sum is therefore strictly increasing to $3/2$ and strictly decreasing afterwards. Dividing by x and by the resulting integral gives a C^1 probability density with all the asserted endpoint properties. For this fixed J, taking ε sufficiently small preserves every strict sign, by the same weighted L^1 convergence argument used in Theorem 6.1. There are $2r+2$ distinct level crossings, and Rolle's theorem gives at least $2r+1$ distinct critical points. □

Formal verification status

This version presents complete mathematical proofs. These proofs have not yet been formalized in a proof assistant. Formal verification is planned; under the author's verification standard, this version is archived as a dated preprint, with active promotion of its results deferred until formal verification is complete.

Declaration of AI assistance

During the preparation of this work, the author used an author-developed local implementation of the Proof Engine infrastructure (development version 2.0) for AI-assisted research. The project-level research framework is described in *Proof Engine 2.0* [3]. See also *Proof Engine Infrastructure* (1.0) [4] for the complementary proof-development framework. The implementation uses replaceable AI-agent and execution-harness components; the present work used OpenAI Codex as its AI-agent component. The author is responsible for the definitions, statements, proofs, citations, code, and exposition.

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